

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 17 Diversity of Archaea

17.1 Multiple Choice Questions

- 1) The microorganisms that are thought to resemble the first cellular forms of life on Earth are
A) H₂-oxidizing hyperthermophiles.
B) acidophilic *Archaea*.
C) the *Nanoarchaeota*.
D) fermentative thermophiles.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

- 2) Which statement is TRUE of methanogenic and extremely halophilic *Archaea*?
A) The methanogens are obligate aerobes; the extreme halophiles are obligate anaerobes.
B) The methanogens are obligate anaerobes; the extreme halophiles are mostly obligate aerobes.
C) Both the methanogens and the extreme halophiles are obligate aerobes.
D) Both the methanogens and the extreme halophiles are obligate anaerobes.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1, 17.2

- 3) *Archaea* are
A) primarily ancient organisms that have not evolved very much in the last billion years.
B) phylogenetically diverse, but functionally restricted to extremophiles and chemolithotrophs.
C) a group of diverse microorganisms that inhabit almost every environment on Earth, including the colon of humans and other mammals.
D) functionally diverse, but restricted to thermophilic or hyperthermophilic environments.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

- 4) Which of the following types of metabolisms is NOT found in *Archaea*?
A) photosynthesis
B) chemolithotrophy
C) chemoorganotrophy
D) aerobic respiration

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.9

5) Some of the largest naturally occurring plasmids occur in the

- A) methanogens.
- B) Archaea lacking cell walls.
- C) extreme halophiles.
- D) genus *Thermococcus*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

6) Most haloarchaea use which of the following for electron donors?

- A) amino and organic acids
- B) carbohydrates
- C) lipids
- D) nucleic acids

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

7) Some strains of haloarchaea exhibit

- A) anoxic growth at the expense of sugar fermentation.
- B) anaerobic respiration.
- C) anoxygenic photosynthesis.
- D) light-driven ATP synthesis under oxygen-limiting conditions.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

8) The cytoplasmic proteins of *Halobacterium* are

- A) highly acidic.
- B) highly basic.
- C) generally neutral.
- D) variable, depending on the species.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

9) Retinal in prokaryotes is part of

- A) photosynthesis.
- B) phototaxis.
- C) light-driven ATP synthesis.
- D) light-driven ATP synthesis and phototaxis.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

10) Methanogens

- A) produce methane as a part of their energy metabolism.
- B) utilize methane as an energy source.
- C) process and store methane to detoxify it.
- D) produce, utilize, and detoxify methane.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

11) What best describes the methanogens?

- A) They are a monophyletic group.
- B) They are phylogenetically similar.
- C) They are a diverse group that is spread across at least 7 orders.
- D) They are functionally but not phylogenetically diverse group.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

12) Most known methanogens are nonhalophilic

- A) psychrophiles.
- B) mesophiles.
- C) thermophiles.
- D) extremophiles.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

13) Which of the following are NOT substrates for methanogens?

- A) acetate and pyruvate
- B) carbon dioxide, carbon monoxide, and formate
- C) glucose and sucrose
- D) methanol and methylamine

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

14) In most salt lakes, which microbial group is the major oxygenic phototroph?

- A) cyanobacteria
- B) *Dunaliella*, a eukaryotic alga
- C) halophilic *Archaea*
- D) purple nonsulfur bacteria

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

15) What special adaptation(s) has/have *Thermoplasma* evolved to survive in hot acidic environments?

A) *Thermoplasma* stabilize their cytoplasmic membrane by incorporation of tetraether lipoglycans in the cytoplasmic membrane.

B) *Thermoplasma* have a reinforced cell wall containing basic amino linkers in the peptidoglycan.

C) *Thermoplasma* excrete hydrogen ions and import hydroxyl ions.

D) *Thermoplasma* are highly motile to avoid destruction.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

16) Approximately _____ of the *Methanocaldococcus jannaschii* genes have no counterparts in known genes from other sequenced organisms.

A) 10%

B) 25%

C) 40%

D) 70%

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

17) Some members of the *Thermoplasmatales* can grow at pH levels below

A) 4.

B) 7.

C) 0.

D) 10.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

18) A chemoorganotroph growing optimally at 55°C and pH _____ in complex media would most likely be a member of the genus

A) *Methanocaldococcus*.

B) *Halobacterium*.

C) *Methanococcus*.

D) *Thermoplasma*.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

19) What characteristic do bacterial mycoplasmas and the archaeal *Thermoplasma* and *Ferroplasma* have in common?

- A) contain plasmids
- B) hyperthermophilic
- C) lack cell walls
- D) relatively large genomes from duplicated genes

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

20) Which genus within Archaea is capable of growth at the hottest temperature recorded of 122°C?

- A) Methanopyrus
- B) Thermoproteus
- C) Thermosphaera
- D) Pyrococcus

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

21) Which archaeal phylum contains many extreme halophiles, methanogens, some marine hyperthermophiles, and *Thermoplasma*?

- A) Crenarchaeota
- B) Euryarchaeota
- C) Nanoarchaeota
- D) Thaumarchaeota

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

22) What makes the metabolism of *Ferroglobus* interesting in the context of evolution?

- A) *Ferroglobus* is an anaerobic iron-oxidizing chemolithotrophic autotroph that may represent a key metabolism that allowed for the oxidation of ferrous iron without molecular oxygen.
- B) *Ferroglobus* can use H₂ or H₂S as electron acceptors in its energy metabolism and is thermophilic, thus representing the earliest hypothesized forms of life.
- C) *Ferroglobus* lacks cell walls and becomes encrusted in oxidized iron, suggesting that it may be an ancestor of other archaeans that lack cell walls.
- D) *Ferroglobus* is both an iron- and nitrate-reducer and is probably the last common ancestor of all iron- and nitrate-reducing organisms.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.4

23) Which of the following bioenergetic processes CANNOT occur in hyperthermophiles?

- A) chemoorganotrophy
- B) chemolithotrophy
- C) photosynthesis
- D) chemoorganotrophy and chemolithotrophy

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

24) Which of the following is TRUE about the crenarchaeal genera *Sulfolobus*?

- A) *Sulfolobus* grows in sulfur-rich hot alkaline springs.
- B) *Sulfolobus* is an anaerobic chemolithotroph.
- C) *Sulfolobus* can grow chemoorganotrophically.
- D) *Sulfolobus* is a gram-positive rod.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.9

25) _____ is a strict anaerobe with irregularly disc-shaped cells and an optimum growth temperature of 105°C. It grows chemolithotrophically on H₂ with S⁰ as an electron acceptor or chemoorganotrophically on complex mixtures of organic compounds.

- A) *Pyrodictium*
- B) *Pyrolobus*
- C) *Pyrobaculum*
- D) *Desulfurococcus*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.10

26) Many archaeal hyperthermophiles are _____; however, *Staphylothermus* is a chemoorganotroph.

- A) chemolithotrophs
- B) mixotrophs
- C) parasites
- D) photoheterotrophs

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.10

27) The factor that most affects the thermostability of a protein is its

- A) highly hydrophobic core regions.
- B) folding pattern.
- C) ionic surface interactions.
- D) number of hydrogen bonds.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

28) One factor that seems to aid in maintaining lipid stability in hyperthermophilic *Archaea* is that dibiphytanyl tetraether lipids form a(n)

- A) covalently bonded monolayer cytoplasmic membrane.
- B) interlocking bilayer cytoplasmic membrane.
- C) interlocking lattice with the cell wall.
- D) pseudomembrane immediately inside the cytoplasmic membrane.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

29) One unusual feature of *Pyrococcus* is that

- A) it incorporates carbon dioxide via the acetyl-CoA pathway.
- B) it is very motile due to a tuft of polar archaeella.
- C) it incorporates carbon dioxide via the reverse citric acid cycle.
- D) it is a thermophilic aerobic chemoorganotroph that uses elemental sulfur.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.4

30) What is the minimum concentration of sodium chloride that an extreme halophile requires for growth?

- A) 1.5 M
- B) 4 M
- C) 6 M
- D) 10 M

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

31) Cellular integrity and cell wall stabilization of the *Halobacterium* are aided by

- A) potassium ions.
- B) chloride ions.
- C) magnesium ions.
- D) sodium ions.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

32) *Halobacterium salinarum* and certain other extreme halophiles carry out light-driven synthesis of ATP under what condition?

- A) low oxygen concentrations
- B) low light
- C) high oxygen concentrations
- D) high carbon dioxide concentrations

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

33) Laboratory isolation of *Nanoarchaeum equitans* requires the presence of

- A) fatty acids.
- B) oxygen.
- C) *Ignicoccus*.
- D) sunlight.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.6

34) Thermostable proteins tend to have additional features that improve thermostability. Which of the following is NOT one of these features?

- A) increased ionic interactions on the protein surfaces
- B) highly hydrophilic cores
- C) decreased tendency of the protein to unfold
- D) highly hydrophobic cores

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

35) What is thought to be the maximum temperature for life to exist?

- A) 125°C
- B) 150°C
- C) 250°C
- D) 500°C

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.11

36) The most common electron donor used among hyperthermophilic *Archaea* is

- A) Fe^{+3}
- B) H_2 .
- C) SO_4 .
- D) S^0 .

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

37) Extremely halophilic *Archaea* often form _____, probably in order to access _____.

- A) stalks / nutrients
- B) intracellular inclusions / elemental sulfur
- C) gas vesicles / oxygen
- D) intracellular membranes / sunlight

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

38) Compatible solutes counteract the tendency of a cell to become

- A) buoyant in high ionic strength environments.
- B) hyperosmotic in low ionic strength environments.
- C) dehydrated in high ionic strength environments.
- D) viscous in low ionic strength environments.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

39) The ribosomes of *Halobacterium* require high _____ levels for stability and activity.

- A) KCl
- B) NaCl
- C) NaSO₄
- D) acidic amino acid

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

40) The cytoplasmic membrane of *Thermoplasma* contains a lipopolysaccharide-like material called lipoglycan that

- A) never contains sugars such as glucose.
- B) forms an unusual and flexible cell wall.
- C) contributes to membrane stability.
- D) may be present or absent dependent upon the temperature.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

41) Crenarchaeotes whose optimal growth temperature is over 80°C are called

- A) hyperthermophiles.
- B) thermophiles.
- C) extreme halophiles.
- D) halophiles.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

42) Hot sulfur-rich environments associated with terrestrial sulfur-rich springs are called

- A) acid-mine drainage.
- B) hydrothermal vents.
- C) solfataras.
- D) terrestrial volcanoes.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.8

43) Proteins known as _____ function to refold partially denatured proteins.

- A) histones
- B) chaperonins
- C) HU proteins
- D) histones and chaperonins

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

44) The topoisomerase called _____ introduces positive supercoils in DNA, which stabilizes DNA and prevents the DNA helix from denaturing at high temperatures.

- A) reverse ligase
- B) ligase
- C) DNA gyrase
- D) reverse DNA gyrase

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

45) Potassium cyclic 2,3-diphosphoglycerate (cDPG) is present at concentrations of 1M or more in the cytoplasm of many hyperthermophilic methanogens. The function of this compound is to

- A) prevent rRNA misfolding.
- B) prevent chemical damage to DNA at high temperatures.
- C) introduce supercoils into DNA.
- D) activate hydrogen gas.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

46) It is difficult to estimate when cyanobacteria appeared on Earth using the evidence from the oxidation state of the Earth's crust because

- A) several ancient anaerobic metabolisms require O₂.
- B) there are multiple forms of anoxygenic photosynthesis.
- C) there are multiple groups of oxygenic photosynthetic bacteria.
- D) several ancient anaerobic metabolisms form Fe³⁺ without requiring O₂.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.4

47) Nitrifying *Thaumarchaeota* can outcompete nitrifying bacteria in acidic environments because *Thaumarchaeota*

A) are well-adapted to high salinity and extreme UV irradiation.

B) are well-adapted to temperature extremes.

C) are well-adapted to very low NH_3 concentrations.

D) have very resistant specialized cells that survive long periods of dehydration and nutrient starvation.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.5

48) The bioenergetics of anaerobic respiring *Archaea* such as those of the *Crenarchaeota* involve _____ from which _____ is synthesized.

A) proton-translocating ATPases / ATP

B) substrate level phosphorylation / ATP

C) sodium translocation / ATP

D) substrate level phosphorylation / GTP

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.8

49) Nitrifying *Archaea* such as *Nitrosopumilus maritimus* use _____ as a sole carbon source and are capable of obtaining sufficient energy by aerobically oxidizing _____.

A) CO_2 / NO

B) sugars / ammonia

C) CO_2 / ammonia

D) organic acids / NO

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.5

50) The small ribosomal rRNA gene sequences of hyperthermophilic *Archaea* and *Bacteria* have relatively high GC content, which

A) increased stability at high temperatures because GC pairs are joined by 3 hydrogen bonds.

B) decreased stability at high temperatures because GC pairs are joined by 2 hydrogen bonds.

C) increased mutation rates at high temperatures compared with other rRNA.

D) considerable variation in their GC content, making this unhelpful as a distinguishing feature between them and non-hyperthermophilic *Archaea* and *Bacteria*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

17.2 True/False Questions

1) The domain *Archaea* is divided into three phyla.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

2) Many hyperthermophiles are chemolithoautotrophs.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

3) *Sulfolobus* can only grow chemoorganotrophically.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.9

4) Most of the *Archaea* are phenotypically quite similar.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

5) Although the domain *Archaea* contains many acidophilic organisms, there are still many others within the domain that grow well in environments with a neutral pH.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.5

6) Some *Archaea* are chemoorganotrophs and use organic compounds as electron donors for energy metabolism.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.8

7) There are many unique archaeal genes, as exhibited by *Methanocaldococcus jannaschii*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

8) In chemolithotrophic *Archaea*, H₂ is a common electron donor.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.13

9) Thus far, only *Archaea* have been found in the salt lakes, while *Bacteria* and *Eukarya* are noticeably absent.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

10) Most halobacteria are motile by means of lophotrichous archaella.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

11) Replacing sodium with potassium can satisfy the requirement for sodium in the halobacteria.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

12) Many halophiles require a high ionic environment for survival.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

13) The light-stimulated proton pump of *Halobacterium salinarum* indirectly functions to pump sodium out of the cell by the activity of a Na^+/H^+ antiport system.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

14) Pseudomurein (pseudopeptidoglycan) is part of the cell wall of some methanogens.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

15) Hyperthermophilic *Archaea* generally have a lower GC ratio in their genomic DNA than in their ribosomal RNA.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

16) The acetoclastic methanogens are mainly classified within one order.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

17) Methanogenesis is coupled with proton motive force formation and ATP synthesis through the activity of ATPases.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.2

18) *Thermoplasma* metabolizes organic compounds leached from hot coal refuse.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

19) Halophilic *Archaea* are the only inhabitants of highly saline environments.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

20) *Thermoplasma* are facultative aerobes, growing either aerobically or anaerobically by sulfur respiration.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

21) *Thermoplasma* contains an extremely small genome complexed with a highly basic DNA-binding protein.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

22) The cytoplasmic membrane of *Picrophilus* is extremely acid impermeable at very low pH.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.3

23) *Archaeoglobus* is a true sulfate reducer that forms a phylogenetically distinct lineage within the *Euryarchaeota*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.4

24) Some crenarchaeotes have growth optima above the boiling point of water (at atmospheric pressure).

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.8

25) With few exceptions, hyperthermophilic crenarchaeotes are obligate aerobes.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.9

26) Some species, mostly classified as Euryarchaeota, have archaeal histones that may contribute to DNA stability as well as having other functions.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

27) When retinal in bacteriorhodopsin becomes excited, it directly catalyzes the formation of ATP.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

28) Archaeal histones assist in maintaining the DNA in a double-stranded form at very high temperatures in species of *Euryarchaeota*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

29) While many *Archaea* are bacilli or cocci, unusual morphologies such as square, flattened and irregular discs have also been identified in certain species.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.10

30) The amino acid composition of thermostable proteins is not particularly unusual, except for a bias for increased levels of amino acids that form α -helices.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

17.3 Essay Questions

1) What makes the lifestyle and genome of *Nanoarchaeum equitans* unique?

Answer: *N. equitans* is the only identified intracellular parasite within the *Archaea*. It lives within the periplasm of *Ignicoccus*. *N. equitans* depends upon its host for the biosynthesis of many essential amino acids and cofactors and may not even be able to make its own ATP. The genome of *N. equitans*, is one of the smallest ever described and is the smallest among *Archaea*. It is also the most dense genome, with over 95% of the genome predicted to encode for proteins. Based on the genes that are missing from the genome, *N. equitans* may depend on *Ignococcus* for both carbon and energy. The key enzymes for DNA repair, DNA replication, transcription, and translation are present in *N. equitans* genome.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.6

2) Describe two mechanisms that *Archaea* use to avoid their DNA melting at temperatures above 100°C.

Answer: Answers will vary. One mechanism involves reverse DNA gyrase which introduces positive supercoils in the DNA and prevents the spontaneous unwinding of the double helix. Another mechanism hyperthermophilic *Archaea* use is the processing of DNA into nucleosome-like structures, which involves positively charged proteins binding to DNA, and the formation of compacted and protected DNA.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

3) Explain why the presence of many *Archaea* has been confirmed only by community sampling of rRNA genes and not by culture in the laboratory. How has metagenomics helped to solve this problem?

Answer: Due to the extreme environmental conditions *Archaea* thrive in, some have been difficult to replicate in the lab. Others might not grow in the lab due to the lack in understanding of essential growth factors or metabolism for a particular archaeon to grow. The plating anomaly is just as much of an issue for culturing *Archaea* from the environment as it is for *Bacteria*. Metagenomics has allowed us to piece together the genomes of *Archaea* from the environment to better predict the metabolism and growth factor requirements of uncultured microbes.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 17.10

4) Why does *Archaeoglobus* represent a metabolically transitional type of organism that bridges the energy-generating processes of H₂S production and methanogenesis?

Answer: *Archaeoglobus* is not generally thought of as a methanogen, yet it does have the ability to produce methane. This functional characteristic and the presence of the specialized coenzymes unique to methanogenesis makes *Archaeoglobus* functionally linked to methanogens.

Archaeoglobus is also a member of the *Euryarchaeota* phylum (as are other methanogens), so the two groups are also linked phylogenetically. *Archaeoglobus* oxidizes H₂ and many organic compounds, reduces SO₄⁻ into H₂S, and the genes required for these respiration processes are present in *Archaeoglobus* as well as many other *Archaea*. The mosaic pattern of metabolisms found in *Archaeoglobus* therefore suggests it might be a transitional type of organism between these different metabolic strategies.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 17.4

5) Hyperthermophilic *Archaea* have been found to contain biphytanyl tetraether lipids. Why are these lipids different from lipids present in non-hyperthermophiles? What special properties and structures do they confer?

Answer: The phytanyl units confer high thermostability in lipids, because they form strong covalent bonds with each other around the lipid (monolayer) membrane. Unlike cells containing a lipid bilayer, biphytanyl tetraether lipids form a lipid monolayer strongly resistant to being pulled apart from heat. The dibiphytanyl tetraether lipids found only in hyperthermophilic *Archaea* help to explain how they are capable of withstanding such high heat while others cannot.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.12

6) *Ignicoccus* has several unique structural features that have lead scientists to propose that it may be the modern descendent of the ancestral cell type that gave rise to the origin of eukaryotic cells. What are the unique features of *Ignicoccus* and why do they support this hypothesis?

Answer: *Ignicoccus* lacks a cell wall, but has both an inner and outer cellular membrane. The space between the two membranes, called the intermediate compartment, is abnormally larger than its cytoplasm and contains membrane-bound vesicles that may function to export substances from the cell. This cell structure resembles some of the membranes in *Eukarya* and supports the idea that *Ignicoccus* may be a modern descendent of the ancestral eukaryote cell. *Ignicoccus* also has a hydrogen- and elemental sulfur-based metabolism and has a temperature optimum of 90°C, thus it also possesses a metabolism that would have been well-adapted for conditions on early earth.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 17.10

7) Detail the mechanism by which halophilic *Archaea* obtain energy from light.

Answer: Halobacteria have the ability to increase the number of bacteriorhodopsin proteins in the cytoplasmic membrane to collect photons of light. The retinal conjugated to bacteriorhodopsin becomes excited, and this process is coupled to the translocation of protons across the cytoplasmic membrane. As protons accumulate on the outer surface of the membrane, the proton motive force is coupled with the activity of ATPase, and ultimately ATP energy is synthesized.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 17.1

8) Where is the thermosome found, and what is its apparent role?

Answer: A thermosome is a complex containing chaperonin proteins that are found within the cells of hyperthermophilic *Archaea*. The function of thermosomes is to facilitate the proper folding of proteins. Thermosomes can also refold protein that have become denatured at high temperatures, thus cells containing thermosomes can survive temperatures beyond the temperature range in which they grow.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 17.12

9) Why does *Halobacterium* pump large quantities of K⁺ cations into its cytoplasm from the environment? What would happen if *Halobacterium* did not do this?

Answer: *Halobacterium* maintains a high intracellular concentration of K⁺ cations so that the high quantities of Na⁺ cations in the environment do not diffuse into the cell. By doing this, a positive water balance is achieved, avoiding death via dehydration.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 17.1

10) Speculate on the possibility of life on other planets. Use representatives of the hyperthermophilic *Archaea* and hydrogen chemolithotrophs as a guide to this discussion.

Answer: Answers will vary, but the environment described should match the temperature tolerances of the *Archaea* thriving in the location. Energetic strategies mentioned should also correspond with compounds present in the environment. For example, hyperthermophilic, chemolithoautotrophic, hydrogen-oxidizing *Archaea* could potentially thrive on a planet containing H₂ in its 120°C atmosphere.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 17.13